Ink Recoverability Protocol (IRP)

Is the ink missing, or just not recovered yet? A CPU-only decision tree with a card per segment.

Diego Lescano · Vesuvius Challenge October 2026 progress prize · MIT · tool: inkcheck (pip install -e .)

- Node A: does a held-out model recover the reference strokes? (band-pass r and stroke recall vs search-matched shift nulls)
- Node R: do two independent references agree? Node N: do negative controls fail?
- Node B: is the scan inside the blur/noise/voxel tolerance of a ~9 µm model? Node C: does a second scan change it?
- Node D: does an unlabelled segment meet a rule written and hashed before the first look?

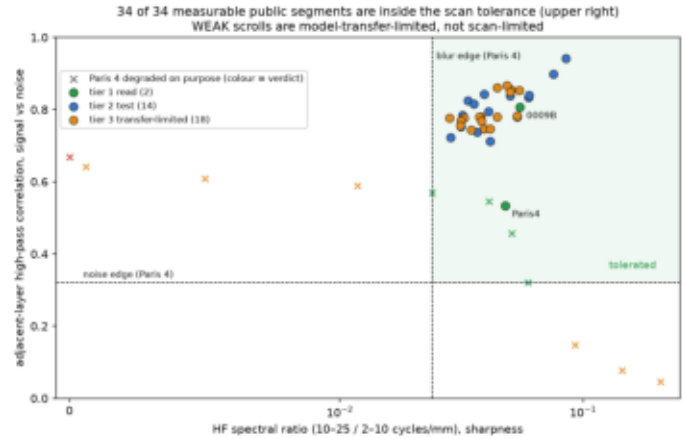
Validation: 10 cards, verdicts produced by `inkcheck report`

PASS	**Paris 4, held out** 5/5 held-out checkpoints PASS (Reader v2, V3, 3 others); scores withheld until the PHercParis4 reservation ends (2027-06-25)
REFS_AGREE (A: PASS)	**PHerc0009B, held-out segment** Reader V3, v2 and dense_native PASS vs team 2.4 µm map (r 0.79 / 0.75 / 0.67) and vs human labels (10/10 strokes)
NO_CONTROL_PASSES	**Negative controls** wrong region, phase-randomised, shuffled tiles: 0/4 PASS (3 FAIL, 1 WEAK with recall at chance)
CONTROL_REPRODUCED	**PHerc0814 validation mask** AUC matches the official evaluation within 0.02 for 4/4 checkpoints
SCROLL_SPECIFIC	**PHerc0343P, held-out segments** Reader v2/V3 WEAK (r 0.18-0.22); 2 refs agree (IR camera letters, ρ 0.90); scan in tolerance; 2.2 µm scan no better
REFS_AGREE (A: WEAK)	**PHerc0841, unseen scroll** Reader V3 / v2 / dense_native WEAK: best r 0.55, recall 0.34 vs 0.15 chance; human labels: r ≤ 0.30, ≤ 1 of 7 strokes
WEAK	**PHerc0172 w082 + w083, held out** Reader V3 / v2 / dense_native WEAK on both: r 0.26-0.51, stroke recall 0.21-0.37 vs 0.16-0.20 chance
WEAK	**PHerc0500P2 gen_00371 + 00377, held out** Reader V3 / v2 / dense_native WEAK on both: r 0.21-0.44, stroke recall 0.10-0.32 vs 0.05-0.07 chance
NO_HELD_OUT	**ScrollGT flagship label (Paris 4), train-exposed** Reader V3 and seed42 PASS, but the segment is in their training data, so IRP excludes it (scores withheld)
NO_CANDIDATE	**PHerc1203 at 2.4 µm, unlabelled** 0/19 zone/model pairs meet the pre-registered candidate rule

Calibration: readable pairs 11/11 PASS (min r 0.58); controls 0/4 PASS (max r 0.15); shift nulls max r 0.16 over 9792 draws (PASS needs r ≥ 0.20 and recall ≥ 0.50). Every check carries 2 mm tile-bootstrap 95% CIs. Thresholds were fixed on synthetic tests before real data; moving the r threshold (0.10-0.30) changes no verdict.

Related work: inkbench (inksweep/inkbench) and ScrollGT (jonmarrs/scrollgt) measure models; IRP decides what to do with a segment (is the scan, the model or the scroll the bottleneck?) and adds pre-registration. villa #2210 (held-out evaluation) and #2318 (face check: writing vs texture) are complementary checks inside villa.

Triage of public segments: where is GPU time worth it?



34 of 34 measurable public segments (10 scrolls) are inside the scan tolerance where a ~9 µm model still reads Paris 4. Where models are WEAK, the scan is not the bottleneck: model transfer is.

Tiers: read: 2 test: 14 transfer-limited: 18 not measured: 1 (table: triage/TRIAGE.md)

Limits: references are model outputs or annotations, not ground truth; the tolerance was measured on Paris 4 only; the 343P IR reference has 8 strokes; scan proxies come from one crop per segment; ScrollGT label placement is ~0.3 mm, so scores against it are lower bounds; data licenses of the Vesuvius open data apply. Full list: README, Limitations.